GLOBAL POLLUTION ANALYSIS AND ENERGY RECOVERY

Project Report

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1. Introduction

Pollution is affected by many factors such as air pollution, water pollution, soil pollution, industrial waste and carbon dioxide emissions. At the same time, some waste can be processed using suitable technologies to recover useful energy. In this project, global pollution data is studied together with waste, energy consumption and energy recovery information. Clustering methods are used to group similar observations, while regression and a neural network are used to study the prediction of energy recovered.

2. Objectives

- To analyse global pollution and energy-related data.
- To check and preprocess the dataset.
- To study pollution levels, industrial waste, CO2 emissions and energy recovery.
- To group similar observations using K-Means and Hierarchical Clustering.
- To use Linear Regression and a feedforward Neural Network to predict energy recovery.
- To compare the prediction results using R^2, MSE and MAE.
- To identify useful observations for pollution management and energy recovery.

3. Dataset Description

The dataset used is Global_Pollution_Analysis.csv. The actual supplied dataset contains 200 observations and 13 original columns. It covers 175 unique countries and contains pollution, waste, energy, population and economic information.

4. Data Preprocessing

The dataset was loaded using Pandas and checked before applying the machine learning methods. There were 0 missing values and 0 duplicate rows in the supplied data. A combined Pollution_Index was created by taking the mean of the air, water and soil pollution indices. Country values were label encoded and the year was kept as a numeric feature. The dataset also already contained Energy_Consumption_Per_Capita, so an estimated total energy consumption feature was calculated from per-capita consumption and population.

5. Exploratory Data Analysis

The main descriptive values from the dataset were: average Air Pollution Index = 180.63, Water Pollution Index = 115.07, Soil Pollution Index = 76.49, Industrial Waste = 52,891.68 tons, CO2 Emissions = 24.88 MT, Energy Recovered = 260.45 GWh and Energy Consumption per Capita = 9.44 MWh. Correlation

analysis showed only weak linear relationships between the selected pollution/waste variables and Energy Recovered. The approximate correlations with Energy Recovered were 0.003 for air pollution, -0.042 for water pollution, 0.064 for soil pollution, -0.161 for industrial waste and 0.025 for CO2 emissions.

6. K-Means Clustering

K-Means clustering was applied after scaling the numerical clustering features. The Elbow Method was used to select the number of clusters. In the final notebook, 3 clusters were used. The clusters were visualized using 2D and 3D plots to see how pollution and energy recovery were distributed. The clustering helps identify observations with similar pollution and energy-related characteristics.

7. Hierarchical Clustering

Agglomerative Hierarchical Clustering was also performed using 3 clusters. A dendrogram was generated to show how observations are joined step by step. The cluster summary in the notebook showed different groups with different average pollution and energy recovery values. This provides another way of checking whether similar patterns are obtained without fixing the clusters in advance in the same way as K-Means.

8. Linear Regression

Linear Regression was used as a simple reference model for predicting Energy Recovered (in GWh). The model was trained using pollution, waste and other selected numerical features. On the test data, the final notebook obtained R^2 = -0.1436, MSE = 23527.4818 and MAE = 134.8496. The negative R^2 indicates that the selected inputs did not explain the test-set variation in energy recovery well.

9. Neural Network

A feedforward neural network was used to model Energy Recovered. The input features included pollution indicators, CO2 emissions, industrial waste and other available numerical variables. The network was trained and its performance was checked using R^2, MSE and MAE. The final run obtained R^2 = -0.3385, MSE = 27538.4980 and MAE = 145.8823. A small hyperparameter comparison was also included in the notebook. For this dataset, the neural network did not improve the test results compared with the Linear Regression reference model.

10. Model Comparison

The final test results were: Linear Regression — R^2 = -0.1436, MSE = 23527.4818, MAE = 134.8496. Neural Network — R^2 = -0.3385, MSE = 27538.4980, MAE = 145.8823. Both models have negative R^2 values on the test set. This means the current features and available sample were not enough to make reliable numerical predictions of energy recovery. The result is useful because it shows that pollution indicators alone should not be treated as a direct measure of recoverable energy.

11. Key Findings

- The dataset contains 200 observations from 175 unique countries.
- There were no missing values or duplicate rows in the supplied dataset.
- Average energy recovered was about 260.45 GWh.
- The selected pollution and waste variables had weak linear correlations with energy recovery.
- K-Means produced 3 groups after applying the Elbow Method.
- Hierarchical clustering also used 3 groups and produced a different but useful view of similarity.
- Linear Regression performed better than the neural network on the final test split.
- The prediction results suggest that more detailed information about waste and energy-conversion

processes is needed.

12. Recommendations

• Improve waste collection and segregation so suitable waste can be used for energy recovery.
• Consider waste-to-energy systems for suitable industrial and municipal waste streams.
• Use pollution and waste data together for environmental planning rather than using a single pollution indicator.
• Increase monitoring of air, water and soil pollution over multiple years.
• For future prediction models, include waste composition, calorific value, conversion efficiency, technology type and operating cost.
• Use a larger dataset with more observations for different countries and years.

13. Limitations

The dataset has only 200 observations and 175 unique countries, so many countries have limited observations. Energy recovery can also depend on technical factors that are not present in the dataset. The prediction models therefore provide a data-analysis result for this dataset and should not be treated as a complete real-world energy-recovery model.

14. Conclusion

This project analysed global pollution and energy-related data using data preprocessing, exploratory analysis, clustering and prediction techniques. K-Means and Hierarchical Clustering were used to group similar observations, while Linear Regression and a feedforward Neural Network were used to predict Energy Recovered. The final test results showed negative R^2 values for both prediction models, which indicates that the available features were not sufficient for accurate prediction. Overall, the project shows how machine learning can be used to study environmental data and also shows why richer technical data is important when estimating energy recovery.

Software Used: Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn and TensorFlow/Keras.